



LECTURE 12

PHP & MySQL

INTRODUCTION

- PHP 5 and later can work with a MySQL database using:
- **MySQLi extension** (the "i" stands for improved)
- **PDO (PHP Data Objects)**
- Earlier versions of PHP used the MySQL extension. However, this extension was deprecated in 2012

Should I Use MySQLi or PDO?

- If you need a short answer, it would be "Whatever you like".
- Both MySQLi and PDO have their advantages:
- PDO will work on 12 different database systems, where as MySQLi will only work with MySQL databases.
- So, if you have to switch your project to use another database, PDO makes the process easy. You only have to change the connection string and a few queries. With MySQLi, you will need to rewrite the entire code - queries included.
- Both are object-oriented, but MySQLi also offers a procedural API.
- Both support Prepared Statements. Prepared Statements protect from SQL injection, and are very important for web application security.

CONNECT DATABASE

Example (MySQLi Object-Oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn = new mysqli($servername,
$username, $password);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn-
>connect_error);
}
echo "Connected successfully";
?>
```

Example (MySQLi Procedural)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn =
mysqli_connect($servername,
$username, $password);

// Check connection
if (!$conn) {
    die("Connection failed: " .
mysqli_connect_error());
}
echo "Connected successfully";
?>
```

CONNECT DATABASE

Example (PDO)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

try {
    $conn = new PDO("mysql:host=$servername;dbname=myDB", $username,
$password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    echo "Connected successfully";
}
catch(PDOException $e)
{
    echo "Connection failed: " . $e->getMessage();
}
?>
```

CREATE DATABASE

MySQLi Object-oriented

```
• <?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn = new mysqli($servername, $username,
$password);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Create database
$sql = "CREATE DATABASE student";
if ($conn->query($sql) === TRUE) {
    echo "Database created successfully";
} else {
    echo "Error creating database: " . $conn->error;
}

$conn->close();
?>
```

MySQLi Procedural

```
• <?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn = mysqli_connect($servername, $username,
$password);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

// Create database
$sql = "CREATE DATABASE student";
if (mysqli_query($conn, $sql)) {
    echo "Database created successfully";
} else {
    echo "Error creating database: " .
mysqli_error($conn);
}

mysqli_close($conn);
?>
```

CREATE DATABASE

PDO

```
• <?php
  $servername = "localhost";
  $username = "username";
  $password = "password";

  try {
    $conn = new PDO("mysql:host=$servername", $username, $password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    $sql = "CREATE DATABASE studentPDO";
    // use exec() because no results are returned
    $conn->exec($sql);
    echo "Database created successfully<br>";
  }
  catch(PDOException $e)
  {
    echo $sql . "<br>" . $e->getMessage();
  }

  $conn = null;
?>
```

CREATE TABLE

MySQLi Object-oriented

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

$tablename = "mytable";
$sql = "CREATE TABLE $tablename (name VARCHAR(25),
email VARCHAR(25))";

if ($conn->query($sql) === TRUE) {
    echo "Table created successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}
//Closes specified connection
$conn->close();
?>
```

MySQLi Procedural

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = mysqli_connect($servername, $username,
$password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

// sql to create table
$tablename = "mytable";
$sql = "CREATE TABLE $tablename (name VARCHAR(25),
email VARCHAR(25))";

if (mysqli_query($conn, $sql)) {
    echo "Table MyGuests created successfully";
} else {
    echo "Error creating table: " . mysqli_error($conn);
}
mysqli_close($conn);
?>
```


CREATE TABLE

PDO

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "studentPDO";

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username, $password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

    // sql to create table
    $tablename = "mytable";
    $sql = "CREATE TABLE $tablename (name VARCHAR(25), email VARCHAR(25))";

    // use exec() because no results are returned
    $conn->exec($sql);
    echo "Table MyGuests created successfully";
}
catch(PDOException $e)
{
    echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>
```

INSERT DATA

```
<html>
<head>
<title>HTML form</title>
</head>
<body>
<form method="POST" action="mysql.php">
<p>Your name: <input type="text" name="name" size="20"></p>
<p>Your e-mail: <input type="text" name="email" size="20"></p>
<p><input type="submit" value="Go!"></p>
</form>
</body>
</html>
```

INSERT DATA

MySQLi Object-oriented

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = new mysqli($servername, $username,
$password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "INSERT INTO myTable (name, email)
VALUES ('$_POST['name']', '$_POST['email']')";

if ($conn->query($sql) === TRUE) {
    echo "New record created successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}
$conn->close();
?>
```

MySQLi Procedural

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = mysqli_connect($servername, $username,
$password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$sql = "INSERT INTO myTable (name, email)
VALUES ('$_POST['name']', '$_POST['email']')";

if (mysqli_query($conn, $sql)) {
    echo "New record created successfully";
} else {
    echo "Error: " . $sql . "<br>" . mysqli_error($conn);
}
mysqli_close($conn);
?>
```

INSERT DATA

PDO

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "studentPDO";

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username, $password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    $sql = "INSERT INTO myTable (name, email)
            VALUES ('$_POST[name]', '$_POST[email'])";
    // use exec() because no results are returned
    $conn->exec($sql);
    echo "New record created successfully";
}
catch(PDOException $e)
{
    echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>
```

SELECT DATA

MySQLi Object-oriented

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = new mysqli($servername, $username,
$password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT name, email FROM myTable";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    // output data of each row
    while($row = $result->fetch_assoc()) {
        echo "Name: " . $row["name"]. " <br>";
        echo "Email: " . $row["email"]. " <br>";
    }
} else {
    echo "0 results";
}
$conn->close();
?>
```

MySQLi Procedural

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = mysqli_connect($servername, $username,
$password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$sql = "SELECT name, email FROM myTable";
$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {
    // output data of each row
    while($row = mysqli_fetch_assoc($result)) {
        echo "Name: " . $row["name"]. " <br>";
        echo "Email: " . $row["email"]. " <br>";
    }
} else {
    echo "0 results";
}
mysqli_close($conn);
?>
```

SELECT DATA

PDO

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "studentPDO";

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username, $password);
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    $stmt = $conn->prepare("SELECT id, firstname, lastname FROM MyGuests");
    $stmt->execute();

    // set the resulting array to associative
    $result = $stmt->setFetchMode(PDO::FETCH_ASSOC);
    foreach(new TableRows(new RecursiveArrayIterator($stmt->fetchAll())) as $k=>$v) {
        echo $v;
    }
}
catch(PDOException $e) {
    echo "Error: " . $e->getMessage();
}
$conn = null;
echo "</table>";
?>
```

UPDATE DATA

MySQLi Object-oriented

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = new mysqli($servername, $username,
$password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "UPDATE myTable SET name='Nur Aisyah' WHERE
email='aisyah@gmail.com'";

if ($conn->query($sql) === TRUE) {
    echo "Record updated successfully";
} else {
    echo "Error updating record: " . $conn->error;
}

$conn->close();
?>
```

MySQLi Procedural

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = mysqli_connect($servername, $username,
$password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$sql = "UPDATE myTable SET name='Nur Aisyah' WHERE
email='aisyah@gmail.com'";

if (mysqli_query($conn, $sql)) {
    echo "Record updated successfully";
} else {
    echo "Error updating record: " . mysqli_error($conn);
}

mysqli_close($conn);
?>
```

UPDATE DATA

PDO

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "studentPDO";

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username, $password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

    $sql = "UPDATE myTable SET name='Nur Aisyah' WHERE email='aisyah@gmail.com'";

    // Prepare statement
    $stmt = $conn->prepare($sql);

    // execute the query
    $stmt->execute();

    // echo a message to say the UPDATE succeeded
    echo $stmt->rowCount() . " records UPDATED successfully";
}
catch(PDOException $e)
{
    echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>
```


DELETE DATA

MySQLi Object-oriented

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = new mysqli($servername, $username,
$password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// sql to delete a record
$sql = "DELETE FROM myTable WHERE
email='aisyah@gmail.com'";

if ($conn->query($sql) === TRUE) {
    echo "Record deleted successfully";
} else {
    echo "Error deleting record: " . $conn->error;
}
$conn->close();
?>
```

MySQLi Procedural

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "student";

// Create connection
$conn = mysqli_connect($servername, $username,
$password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " .
mysqli_connect_error());
}

// sql to delete a record
$sql = "DELETE FROM myTable WHERE
email='aisyah@gmail.com'";

if (mysqli_query($conn, $sql)) {
    echo "Record deleted successfully";
} else {
    echo "Error deleting record: " .
mysqli_error($conn);
}
mysqli_close($conn);
?>
```

DELETE DATA

PDO

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "studentPDO";

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username,
$password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

    // sql to delete a record
    $sql = "DELETE FROM myTable WHERE email='aisyah@gmail.com'";

    // use exec() because no results are returned
    $conn->exec($sql);
    echo "Record deleted successfully";
}
catch(PDOException $e)
{
    echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>
```